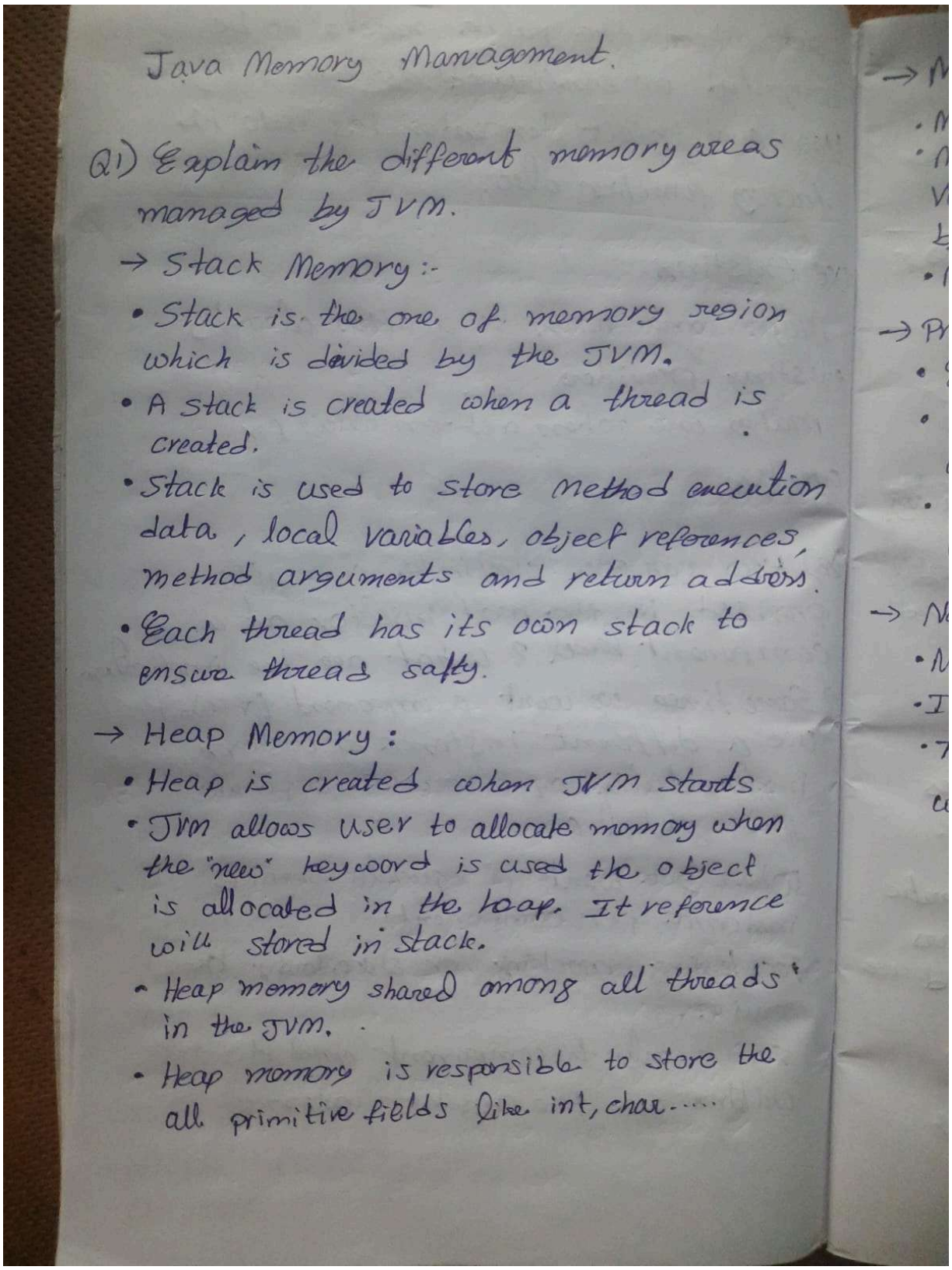
1. Explain the different memory areas managed by JVM.



Java Memory Management.

Q1) Explain the different memory areas managed by JVM.

→ Stack Memory :-

- Stack is the one of memory region which is divided by the JVM.
- A stack is created when a thread is created.
- Stack is used to store method execution data, local variables, object references, method arguments and return address.
- Each thread has its own stack to ensure thread safty.

→ Heap Memory :

- Heap is created when JVM starts
- JVM allows user to allocate momory when the "new" keyword is used the object is allocated in the heap. It reference will stored in stack.
- Heap memory shared among all threads in the JVM.
- Heap memory is responsible to store the all primitive fields like int, char.....

→ Method Area :-

- Method Area also created when JVM started.
- Method Area is used to store ~~local~~ static variables, class level information, method bytecodes and constant pools.
- Method area is basically a part of the heap.

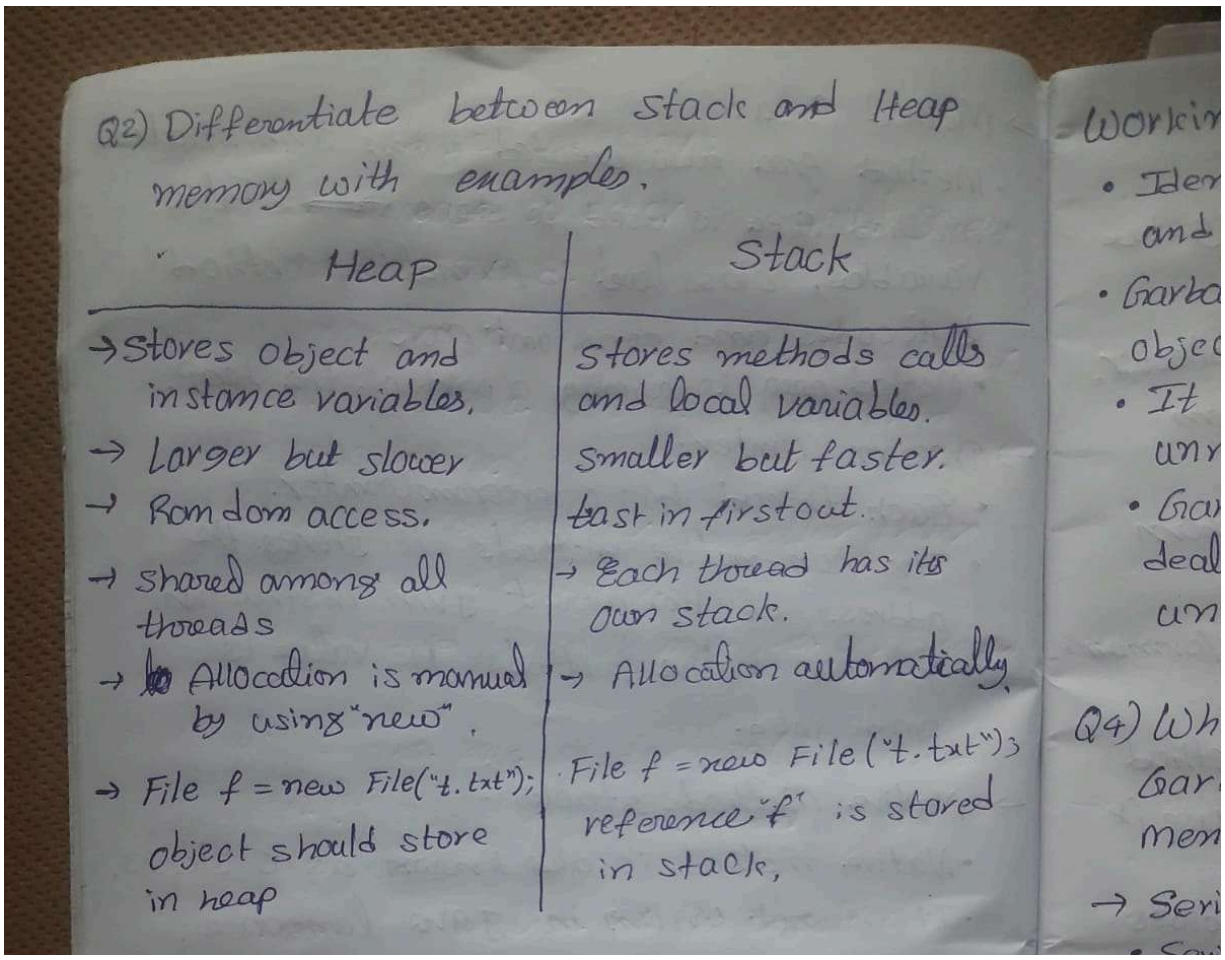
→ Program Counter :-

- Each thread has a program Counter.
- For non-native methods, it stores the address of the current JVM instruction.
- For non-native methods, PC value is undefined.

→ Native Method stacks :-

- Native method is also known as C stacks.
- It is not written in Java language.
- This memory is allocated for each thread when it is created.

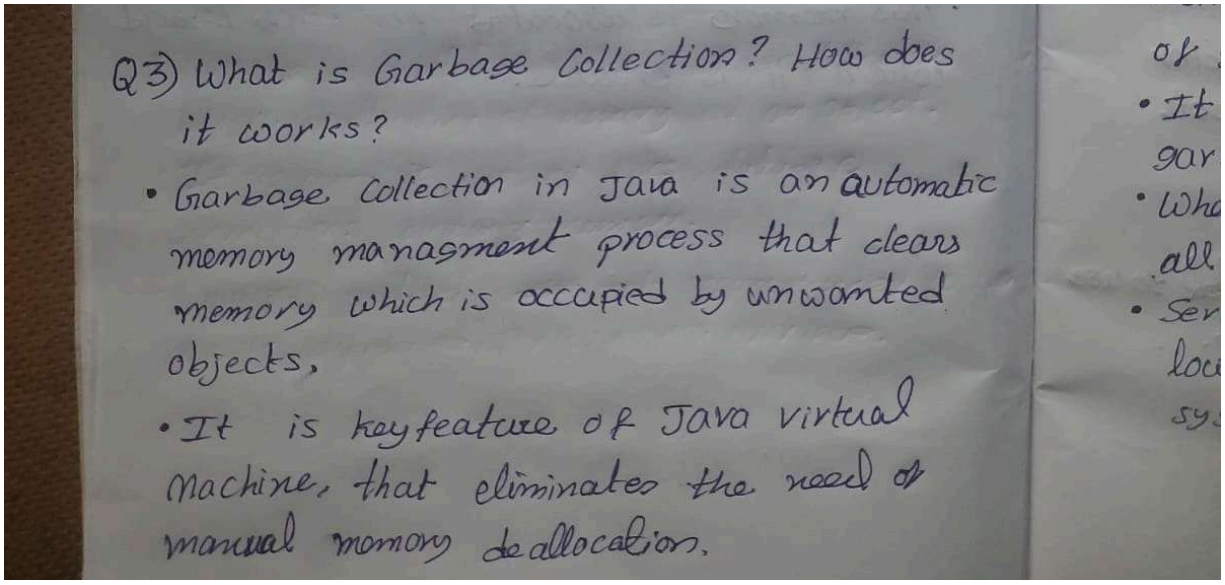
2. Differentiate between Stack and Heap memory with examples.



Q2) Differentiate between stack and Heap memory with examples.

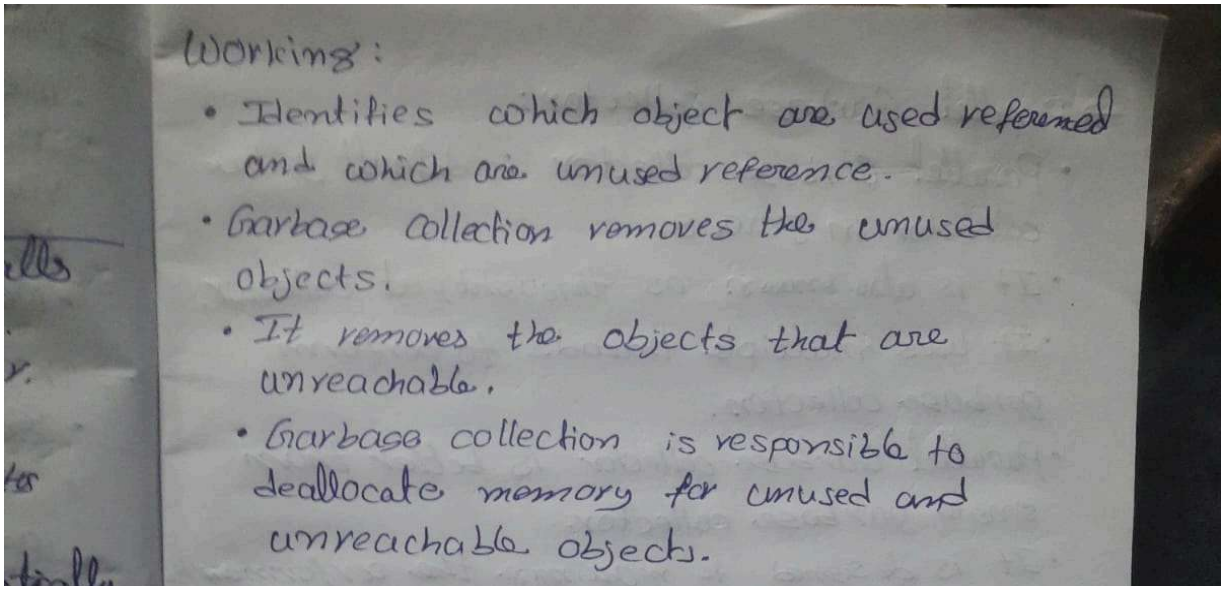
| Heap | Stack |
|---|---|
| → Stores object and instance variables. | Stores methods calls and local variables. |
| → Larger but slower | smaller but faster. |
| → Random access. | last in firstout. |
| → shared among all threads | → Each thread has its own stack. |
| → Allocation is manual by using "new". | → Allocation automatically. |
| → File f = new File("t.txt"); object should store in heap | File f = new File("t.txt"); reference "f" is stored in stack, |

3. What is Garbage Collection? How does it work in Java?



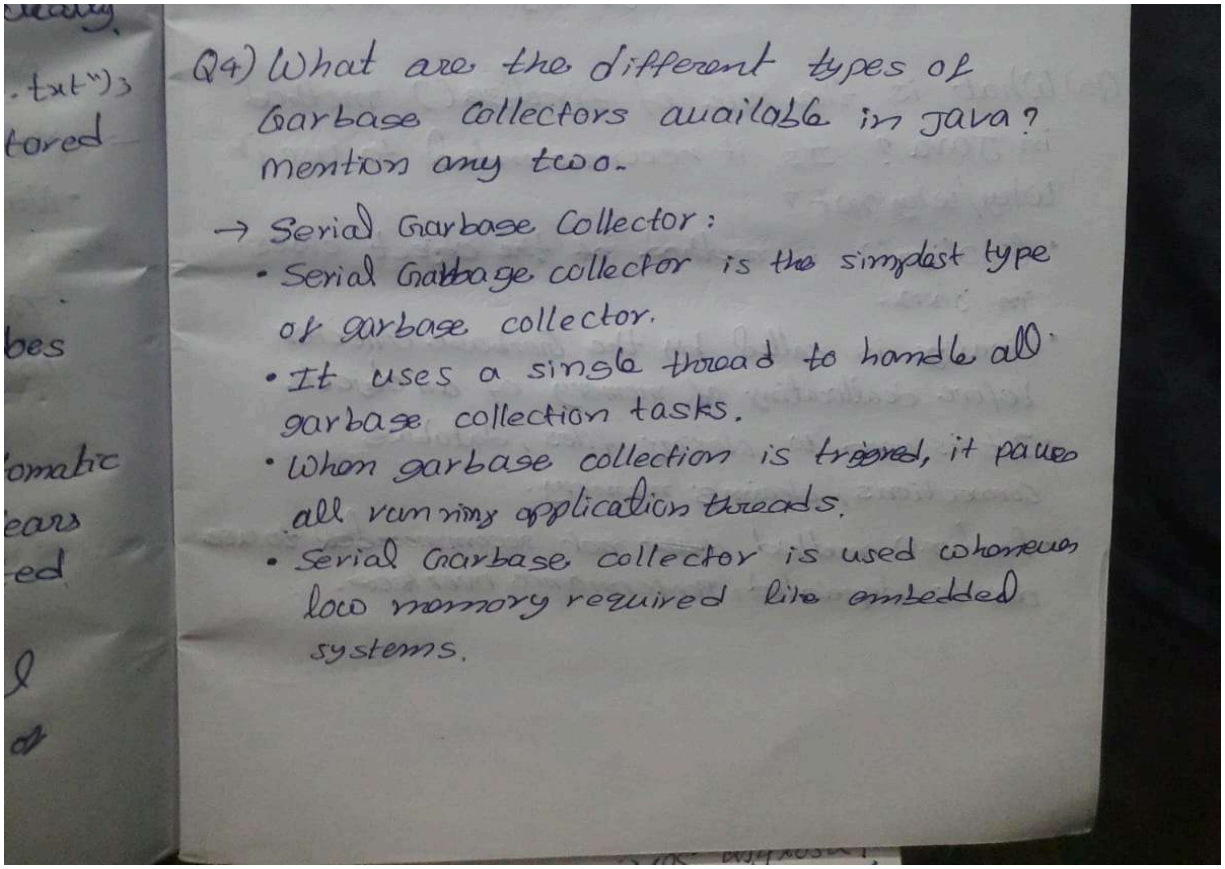
Q3) What is Garbage Collection? How does it works?

- Garbage Collection in Java is an automatic memory managment process that clears memory which is occupied by unwanted objects.
- It is key feature of Java virtual machine, that eliminates the need of manual memory deallocation.


Working:

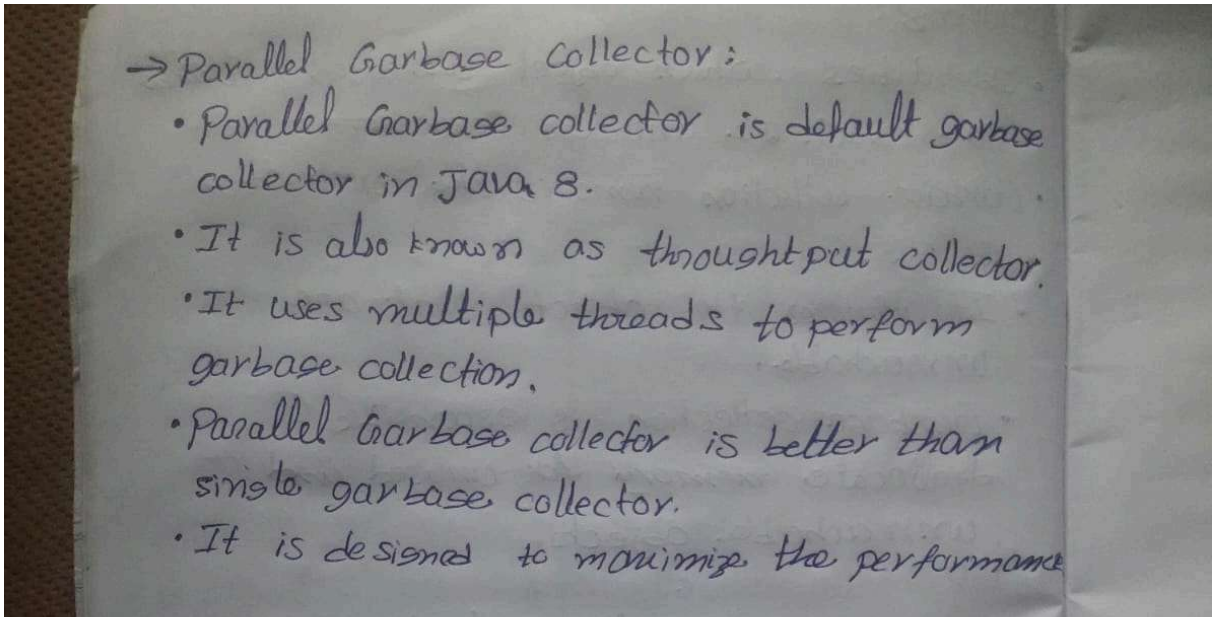
- Identifies which object are used referred and which are unused reference.
- Garbage collection removes the unused objects.
- It removes the objects that are unreachable.
- Garbage collection is responsible to deallocate memory for unused and unreachable objects.

4. What are the different types of Garbage Collectors available in Java? Mention any two.


Q4) What are the different types of Garbage Collectors available in Java? Mention any two.

→ Serial Garbage Collector:

- Serial Garbage collector is the simplest type of garbage collector.
- It uses a single thread to handle all garbage collection tasks.
- When garbage collection is triggered, it pauses all running application threads.
- Serial Garbage collector is used whenever low memory required like embedded systems.




5. What is the role of the finalize() method in Java? Is it recommended to use it? Why/Why not?